Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part –June 26th 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please write a short pseudo-code showing how the **map** parallel pattern could be implemented in OpenMP. (2)

B. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=1; i<100; i++)
        for (j=i; j<100; j++)
            b[i][j] += f(a[i][j]);
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the parallel reduce pattern: assumptions, complexity, and example. (2)

B. Please describe how the Split parallel pattern works. How is it implemented? (2)

Exercise n. 3

Answer the following questions about shared memory parallel programming and briefly explain (without an explanation, the answer will be considered invalid)

A. What error will arise compiling the following code snippet? (1.5)

```
#pragma omp parallel for private(x) reduction(+:sum) default(none)
for (i=1;i<= num_steps; i++)
{
x = (i-0.5)*step;
sum = sum + 4.0/(1.0+x*x);
}
```

B. What happens when a variable is declared as firstprivate in an OpenMP pragma? (1)

C. Why is cancellation in OpenMP described as "best effort" cancellation? (1.5)

Exercise n. 4

Answer the following questions about MPI and briefly explain (without an explanation, the answer will be considered invalid)

A. Why should I use the `MPI_Probe` function? (1) **This question can be skipped if you passed the challenge.**

B. Provide an example(s) of how the MPI `tag` and `rank` can be used in a program (1)

C. What are communicators in MPI? Why should I use more than one? (2)